INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Mathematics

Engineering Mathematics – I (CSE-1001)

Mini-Test, 1 October 2026 — Solutions and Marking Scheme

B.Tech Semester I · CS (AI), **Section 4** · Maximum Marks 20

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1 (A)(a) [1]

Give an example of $f : \mathbb{R} \to \mathbb{R}$ which is continuous on $\mathbb{R}$ and differentiable at every point except at each integer.

$$\boxed{f(x) = |\sin \pi x|}$$

Justification. $\sin \pi x$ is continuous on $\mathbb{R}$, hence so is $|\sin \pi x|$. Its zeros are exactly the integers, and they are *simple*: $\dfrac{d}{dx} \sin \pi x = \pi \cos \pi x = \pm\pi \neq 0$ at every integer, so $\sin \pi x$ changes sign there. Since $|u|$ fails to be differentiable precisely at a sign-changing zero of u, the points of failure are exactly $x \in \mathbb{Z}$. Between consecutive integers $\sin \pi x$ keeps one sign, so $|\sin \pi x| = \pm \sin \pi x$ there and is differentiable.

At an integer n the one-sided derivatives are $+\pi$ and $-\pi$; at a half-integer both are 0, so f is differentiable there, as required.

Also acceptable. $f(x) = t(1-t)$ where $t = x - \lfloor x \rfloor$ (a parabolic arch on each interval): continuous, with one-sided derivatives $+1$ and -1 at every integer and 0 at half-integers.

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1 (A)(b) [1]

$f(x) = x \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Determine, with justification, whether f is differentiable at $x = 0$.

f is NOT differentiable at $x = 0$.

The difference quotient is

$$\frac{f(h) - f(0)}{h} = \frac{h \sin(1/h) - 0}{h} = \sin\left(\frac{1}{h}\right),$$

and this has *no limit* as $h \to 0$. Two subsequences settle on different values:

$$h_n = \frac{1}{2n\pi} \implies \sin \frac{1}{h_n} = 0, \qquad h_n = \frac{1}{\left(2n + \frac{1}{2}\right)\pi} \implies \sin \frac{1}{h_n} = 1.$$

Since the limit does not exist, $f'(0)$ does not exist.

Remark (not required). f *is* continuous at 0, since $|x \sin(1/x)| \leq |x| \to 0$. It is the division by h that destroys differentiability: the factor x is consumed by the quotient and nothing remains to damp the oscillation.

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1 (B)(a) [2]

$f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$, and f is differentiable at $x = 0$. Prove that f is differentiable at every $x \in \mathbb{R}$.

Step 1 — the value of $f(0)$. Putting $x = y = 0$,

$$f(0) = f(0)^2 \implies f(0)\big(f(0) - 1\big) = 0 \implies f(0) = 0 \ \text{ or } \ f(0) = 1.$$

Step 2 — the case $f(0) = 0$. Then for every x,

$$f(x) = f(x + 0) = f(x)f(0) = 0,$$

so $f \equiv 0$, which is differentiable on all of $\mathbb{R}$ with $f' \equiv 0$. The conclusion holds.

Step 3 — the case $f(0) = 1$. Fix any $x \in \mathbb{R}$. Using the functional equation on $f(x + h)$,

$$\frac{f(x+h) - f(x)}{h} = \frac{f(x)f(h) - f(x)}{h} = f(x) \cdot \frac{f(h) - 1}{h} = f(x) \cdot \frac{f(h) - f(0)}{h}.$$

As $h \to 0$ the last quotient tends to $f'(0)$, which exists by hypothesis. Hence the limit exists for every x, and

$$\boxed{f'(x) = f'(0)\, f(x)}$$

so f is differentiable at every $x \in \mathbb{R}$. ■

The idea. The functional equation *translates* the difference quotient at an arbitrary x back to the difference quotient at 0. One hypothesis, at one point, does all the work.

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1 (B)(b) [1]

Find k for which $f(x) = x^2$ $(x \le 1)$, $f(x) = kx$ $(x > 1)$ is continuous at $x = 1$. With that k, is f differentiable at $x = 1$?

Continuity. $\lim\limits_{x \to 1^-} f(x) = 1$, $\lim\limits_{x \to 1^+} f(x) = k$, and $f(1) = 1$. These agree precisely when

$$\boxed{k = 1}$$

Differentiability. With $k = 1$, the one-sided derivatives at $x = 1$ are

$$f'(1^-) = 2x\big|_{x=1} = 2, \qquad f'(1^+) = k = 1.$$

They are unequal, so f is **not** differentiable at $x = 1$. The graph has a corner there: continuity was achieved by matching values, but the slopes still disagree.

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2 (A)(a) [2]

Give, with justification, one example of a pair of sequences (a_n), (b_n) for each of the four prescribed behaviours.

	Example	**Justification**
(i)	$a_n = n,\ \ b_n = -n$	Both are unbounded, hence divergent. $a_n + b_n = 0$ for every n, so the sum converges (to 0).
(ii)	$a_n = b_n = (-1)^n$	Each has subsequential limits $+1$ and -1, so neither converges. $a_n b_n = (-1)^{2n} = 1$ for every n, so the product converges to $1 \neq 0$.
(iii)	$a_n = 0,\ \ b_n = (-1)^n$	$a_n \to 0$ (convergent); b_n diverges as in (ii); $a_n b_n = 0 \to 0$.
(iv)	$a_n = \dfrac{5}{n},\ \ b_n = n$	$a_n \to 0$ and $b_n \to \infty$, while $a_n b_n = 5$ for every n, so $a_n b_n \to 5$.

What the question is testing. The sum and product rules for limits require *both* limits to exist. These four parts show, concretely, what survives when that hypothesis fails — and (iv) is the indeterminate form $0 \cdot \infty$ made explicit.

2 (A)(b) [1]

Examine the convergence of $a_n = (1^n + 2^n + 3^n)^{1/n}$ *and determine its limit.*

The largest of the three terms is 3^n, so

$$3^n \ \leq\ 1^n + 2^n + 3^n \ \leq\ 3 \cdot 3^n.$$

Taking n-th roots (an increasing operation on positive numbers),

$$3 \ \leq\ a_n \ \leq\ 3 \cdot 3^{1/n}.$$

Since $3^{1/n} \to 1$, both bounds tend to 3. By the **Sandwich Theorem** the sequence **converges**, with

$$\boxed{\lim_{n\to\infty} a_n = 3}$$

Numerical check (for the marker). $a_{20} = 3.0000451$, $a_{100} = 3.0000000$; the upper bound $3 \cdot 3^{1/n}$ is 3.0331 at $n = 100$.

Generalisation (worth mentioning in class, not a mark): for positive constants $c_1, \ldots, c_k$,

$$(c_1^n + c_2^n + \cdots + c_k^n)^{1/n} \ \longrightarrow\ \max_{1\leq i\leq k} c_i,$$

the largest base always wins.

2 (B) [2]

Evaluate $\displaystyle\lim_{n\to\infty} \left(\frac{1}{n^2+1} + \frac{2}{n^2+2} + \cdots + \frac{n}{n^2+n} \right)$.

Write $S_n = \displaystyle\sum_{k=1}^{n} \frac{k}{n^2+k}$. The denominators increase with k, running from n^2+1 up to n^2+n.

Replacing every denominator by the largest, and then by the smallest, and using $\sum_{k=1}^{n} k = \dfrac{n(n+1)}{2}$:

$$\frac{n(n+1)/2}{n^2+n} \;\le\; S_n \;\le\; \frac{n(n+1)/2}{n^2+1}.$$

The left-hand bound simplifies exactly:

$$\frac{n(n+1)/2}{n(n+1)} = \frac{1}{2},$$

and the right-hand bound is $\dfrac{n^2+n}{2(n^2+1)} \to \dfrac{1}{2}$. By the **Sandwich Theorem**,

$$\boxed{\lim_{n\to\infty} S_n = \frac{1}{2}}$$

Numerical check (for the marker). At $n = 10^6$: $0.50000000 \le 0.50000017 \le 0.50000050$.

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3 (A) [3]

(i) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ [1]

Converges, by the integral test. Note first that the *ratio test is useless here*: the ratio tends to 1 (numerically 0.99999 at $n = 10^5$).

Take $f(x) = \dfrac{1}{x(\ln x)^2}$, which is positive, continuous and decreasing on $[2, \infty)$. Substituting $u = \ln x$,

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^{\infty} \frac{du}{u^2} = \left[-\frac{1}{u}\right]_{\ln 2}^{\infty} = \frac{1}{\ln 2} < \infty.$$

The integral converges, hence so does the series. (Partial sums level off near 2.04.)

Contrast worth noting. $\sum \frac{1}{n \ln n}$ *diverges* — the square in the denominator is the entire difference.

(ii) $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n^2+1} - n\right)$ [1]

Diverges, by limit comparison. Rationalise first:

$$\sqrt{n^2+1} - n = \frac{\left(\sqrt{n^2+1} - n\right)\left(\sqrt{n^2+1} + n\right)}{\sqrt{n^2+1} + n} = \frac{1}{\sqrt{n^2+1} + n}.$$

Comparing with $\sum \frac{1}{n}$:

$$\lim_{n\to\infty} \frac{1/\left(\sqrt{n^2+1} + n\right)}{1/n} = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+1} + n} = \frac{1}{2},$$

finite and non-zero, so the two series behave alike. Since $\sum \frac{1}{n}$ diverges, so does the given series. (Partial sums reach 7.09 at $N = 10^6$ and continue to grow.)

(iii) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n^2+1}$ [1]

Conditionally convergent.

Convergence, by Leibniz's test. With $a_n = \dfrac{n}{n^2+1}$: $a_n \to 0$; and for $f(x) = \dfrac{x}{x^2+1}$,

$$f'(x) = \frac{1-x^2}{(1+x^2)^2} < 0 \quad \text{for } x > 1,$$

so (a_n) is decreasing. Both conditions hold, and the series converges (numerically to ≈ 0.2696).

Not absolutely. $\dfrac{n}{n^2+1} \sim \dfrac{1}{n}$, and by limit comparison with $\sum \frac{1}{n}$ the series $\sum |a_n|$ **diverges**.

Convergent but not absolutely convergent: **conditionally convergent**.

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3 (B) [2]

Discuss the convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{(3x-1)^n}{n\,4^n}$*, stating the interval of convergence with both endpoints tested separately.*

Step 1 — the radius. Factor the 3 out first: $3x - 1 = 3\left(x - \frac{1}{3}\right)$. By the ratio test,

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{|3x-1|}{4} \cdot \frac{n}{n+1} \longrightarrow \frac{3\left|x-\frac{1}{3}\right|}{4},$$

which is < 1 exactly when $\left|x - \frac{1}{3}\right| < \dfrac{4}{3}$. So the centre is $x = \frac{1}{3}$ and the radius is $R = \frac{4}{3}$, giving the open interval $\left(-1,\ \frac{5}{3}\right)$.

Step 2 — the endpoints.

- $x = \frac{5}{3}$: then $3x - 1 = 4$, and the series becomes $\sum \dfrac{4^n}{n\,4^n} = \sum \dfrac{1}{n}$, the harmonic series — **diverges**.
- $x = -1$: then $3x - 1 = -4$, and the series becomes $\sum \dfrac{(-4)^n}{n\,4^n} = \sum \dfrac{(-1)^n}{n}$ — **converges** by Leibniz's test.

$$\boxed{\text{Interval of convergence } = \left[-1,\ \tfrac{5}{3}\right)}$$

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4 (A) [3]

Obtain the linear and quadratic approximation to $f(x) = \sqrt{x}$ *about* $a = 1$*, estimate* $\sqrt{1.1}$*, and obtain an upper bound for the error in each case using the Remainder Estimation Theorem.*

Derivatives at the base point.

$$f(x) = x^{1/2}, \quad f'(x) = \tfrac{1}{2}x^{-1/2}, \quad f''(x) = -\tfrac{1}{4}x^{-3/2}, \quad f'''(x) = \tfrac{3}{8}x^{-5/2};$$
$$f(1) = 1, \qquad f'(1) = \tfrac{1}{2}, \qquad f''(1) = -\tfrac{1}{4}.$$

The two approximations.

$$L(x) = 1 + \tfrac{1}{2}(x-1), \qquad Q(x) = 1 + \tfrac{1}{2}(x-1) - \tfrac{1}{8}(x-1)^2 \quad \left[\text{since } \tfrac{f''(1)}{2!} = -\tfrac{1}{8}\right].$$

With $h = x - 1 = 0.1$:

$$L(1.1) = 1 + 0.05 = \mathbf{1.05000000}, \qquad Q(1.1) = 1.05 - \frac{0.01}{8} = \mathbf{1.04875000}.$$

Error bounds. On [1, 1.1] the relevant derivatives are largest at $x = 1$, so $M_2 = \max|f''| = \frac{1}{4}$ and $M_3 = \max|f'''| = \frac{3}{8}$. By the Remainder Estimation Theorem,

$$|R_1(1.1)| \le \frac{M_2}{2!}h^2 = \frac{1/4}{2}(0.1)^2 = \mathbf{1.25 \times 10^{-3}},$$

$$|R_2(1.1)| \le \frac{M_3}{3!}h^3 = \frac{3/8}{6}(0.1)^3 = \mathbf{6.25 \times 10^{-5}}.$$

Check for the marker only (the student is *not* given the true value). $\sqrt{1.1} = 1.04880885$, so the actual errors are 1.191×10^{-3} and 5.885×10^{-5}. Both sit just below their bounds — the quadratic bound is only about 6% above the true error.

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4 (B) [2]

Obtain the Maclaurin series of $\cos x$ *and state the values of* x *for which it is valid.*

Derivatives at 0. With $f(x) = \cos x$,

$$f(0) = 1, \quad f'(0) = -\sin 0 = 0, \quad f''(0) = -\cos 0 = -1, \quad f'''(0) = \sin 0 = 0, \quad f^{(4)}(0) = \cos 0 = 1,$$

and the pattern 1, 0, −1, 0 repeats with period 4. Substituting into $\sum_{n\ge 0} \frac{f^{(n)}(0)}{n!}x^n$, the odd-order terms vanish and

$$\boxed{\cos x \;=\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \;=\; \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}}$$

Validity: for every $x \in \mathbb{R}$. The series represents $\cos x$ exactly when the Taylor remainder tends to 0. Every derivative of $\cos x$ is $\pm\sin x$ or $\pm\cos x$, so $\left|f^{(n+1)}(c)\right| \le 1$ for all c and all n. By the Remainder Estimation Theorem,

$$|R_n(x)| = \left|\frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1}\right| \le \frac{|x|^{n+1}}{(n+1)!} \;\longrightarrow\; 0 \qquad (n \to \infty)$$

for every fixed x, since factorial growth outpaces any fixed power. The equality therefore holds on all of $\mathbb{R}$.